CodeX Quality Assurance Test Plan

Release v2.4.0 — Super Admin Security, Rate Limiting, Certificate Engine & SEO

This document serves as the formal test verification guide for the QA testing team. It details test scenarios, step-by-step reproduction instructions, expected outcomes, and acceptance criteria for all features in branch 'feat/super-admin-and-production-certificates'.

TEST ENVIRONMENT & EXECUTION METADATA

Target Branch:	feat/super-admin-and-production-certificates	**Target OS / Platform:**	Windows / Linux / Docker Node v24
Frontend URL:	http://localhost:5173	**Backend API URL:**	http://localhost:5000 /api/v1
Default Admin Role:	Super Administrator (Root Console)	**Rate Limit Baseline:**	Tiered Token Bucket (HTTP 429)
Brand Tagline:	Coding the Future, Today.	**QA Lead / Tester:**	Assigned QA Engineer (Sign-off on P.5)

1. Hardened Two-Step Admin Authentication & MFA Telemetry

Verifies timing-attack mitigation, single-input OTP UX, live attempts telemetry, and lockout rules.

TC-AUTH-01 Step 1: Admin Credentials Validation & Timing Attack Defense

[ ] PASS [ ] FAIL

Route / Target: `/admin/login (Step 1 Form)`

Execution Steps:

- Navigate to http://localhost:5173/admin/login in an incognito window.
- Submit an un-registered email address (e.g., non-existent@codex.com) with an arbitrary password.
- Submit an invalid password for a registered administrator account.

Expected Result:

System returns uniform response times without leaking whether the account exists (via pre-computed dummy bcrypt hash). UI displays clear, non-revealing error message. Password field remains masked.

TC-AUTH-02 Step 2: Clean 6-Digit MFA Input, Typing & Clipboard Auto-Paste

[ ] PASS [ ] FAIL

Route / Target: `/admin/login (Step 2 OTP Transition)`

Execution Steps:

- Log in with valid admin credentials. Verify smooth transition to Step 2 OTP card.
- Confirm the screen displays a single centered input field: [KeyRound | ••••••].
- Type a 6-digit OTP code directly using numeric keyboard. Confirm digits advance cleanly.
- Clear field, copy an OTP (e.g. '123456') from clipboard and paste with Ctrl+V.

Expected Result:

Pasted text is auto-trimmed, formatted, and triggers automatic form verification without manual enter/click. Clunky 6-box input and duplicate card stacking are completely gone.

TC-AUTH-03 Live Remaining Attempt Telemetry & Inline Change Email Flow

[ ] PASS [ ] FAIL

Route / Target: `/admin/login (Step 2 OTP Card)`

Execution Steps:

- Enter an incorrect 6-digit code on Step 2 and submit.
- Inspect the dynamic counter beside the label and the banner: 'X of 5 attempts remaining'.
- Click the inline '• Change email' button.

Expected Result:

System smoothly transitions back to Step 1 email/password card without page reload or stale session tokens. Counter persists remaining quota accurately.

TC-AUTH-04 **Brute-Force Lockout Defense (5 Consecutive Failed Attempts)** [ ] PASS [ ] FAIL

Route / Target: `/admin/login (Step 2 OTP Security Threshold)`

Execution Steps:

- Enter 5 consecutive invalid OTP codes on the active admin session.
- Observe error state on the 5th failed submission.
- Attempt to submit another OTP code immediately.

Expected Result:

Account/session is placed in a 15-minute brute-force lockout. System blocks further submissions, surfaces a high-severity red lockout banner, and invalidates the pending OTP secret.

2. Enterprise Rate Limiting & HTTP 429 Cooldown Banners

Verifies structured HTTP 429 telemetry, CORS exposed headers, and live frontend countdown timers.

TC-RATE-01 **Rapid Request Throttling & HTTP 429 Payload Compliance** [ ] PASS [ ] FAIL

Route / Target: `Backend Endpoints (Auth / Contact / Verify API)`

Execution Steps:

- Simulate rapid requests to an endpoint (e.g., repeatedly submit contact form or invalid logins).
- Inspect the Network tab in Chrome DevTools upon receiving HTTP 429 (Too Many Requests).
- Verify response body contains: { success: false, retryAfter: <sec>, resetTime: <ISO>, remaining: 0 }.
- Check Response Headers for: RateLimit-Limit, RateLimit-Remaining, RateLimit-Reset, Retry-After.

Expected Result:

All headers are cleanly exposed via CORS (configured in app.js). Payload is strictly structured JSON; no raw text or broken crash dumps.

TC-RATE-02 **Frontend Live Countdown Toast / Cooldown Alert in GlobalMessage** [ ] PASS [ ] FAIL

Route / Target: `Frontend UI GlobalMessage Component`

Execution Steps:

- Trigger an HTTP 429 rate limit error through the UI.
- Observe the top-right / screen-top notification alert.
- Watch the countdown timer tick down second-by-second (e.g., 'Please wait 59s, 58s...').

Expected Result:

Axios interceptor catches the 429 status code and displays a live ticking cooldown banner. Submit button is disabled until cooldown timer reaches 0s.

TC-RATE-03 **Admin Settings Security Policies & Rate Limits Overview** [ ] PASS [ ] FAIL

Route / Target: `/admin/settings (Security Policies Card)`

Execution Steps:

- Log in as Administrator and navigate to http://localhost:5173/admin/settings.
- Locate the 'Security Policies & Rate Limits' status card.
- Verify policy tiers are displayed: Auth Limiter (5 req/15m), API Limiter (100 req/15m).

Expected Result:

Security card renders active policies, window duration, max thresholds, and current enforcement status clearly.

3. Certificate Lifecycle, Rejection Deletion & Edit Synchronization

Verifies certificate generation, dispatch statuses, and cascading deletion on rejection or edits.

TC-CERT-01 **Automatic Certificate Deletion on Registration/Certificate Rejection** [ ] PASS [ ] FAIL

Route / Target: `/admin/certificates & /admin/registrations`

Execution Steps:

- Locate an issued certificate in /admin/certificates with an assigned Certificate ID.
- Click 'Delete' or navigate to registrations and reject the participant's certificate.
- Confirm the deletion prompt.
- Open the verification URL: http://localhost:5173/verify-certificate/<certificateId> in browser.

Expected Result:

Certificate is deleted from database and Cloudinary storage. Verification URL immediately returns 'Certificate Not Found / Revoked' with a red warning badge.

TC-CERT-02 **Edit Synchronization & Cryptographic Re-generation** [ ] PASS [ ] FAIL

Route / Target: `/admin/certificates (Edit Modal)`

Execution Steps:

- Select an existing issued certificate and click 'Edit'.
- Change the Student Name (e.g. correct a typo) and Position (e.g. from 'Participant' to 'Winner').
- Save changes and wait for batch synchronization.
- Reload the public verification link and inspect the rendered certificate canvas.

Expected Result:

Metadata updates instantly. The rendered canvas immediately displays updated name and position. Old cache is invalidated and previous batch counters update cleanly.

TC-CERT-03 **Delivery Status Tracking (Pending Dispatch, Dispatched, Delivered, Failed)** [ ] PASS [ ] FAIL

Route / Target: `/admin/certificates (Certificate Table)`

Execution Steps:

- Inspect the 'Delivery Status' column for newly issued certificates.
- Verify initial status displays as 'Pending Dispatch' with an amber pill.
- Trigger 'Resend' or batch dispatch job.
- Verify state transitions to 'Dispatched' or 'Delivered' upon email confirmation.

Expected Result:

Delivery statuses update reactively without requiring full page refresh. Actions menu allows targeted Resend or Delete for any specific recipient.

4. Bulk CSV Import & Export Handling

Verifies validation rules, duplicate email filtering, batch creation, and single-click CSV export.

TC-CSV-01 **Valid CSV Upload & Batch Certificate Creation** [ ] PASS [ ] FAIL

Route / Target: `/admin/certificates (Bulk CSV Import)`

Execution Steps:

- Prepare a test CSV file containing 10 rows: Student Name, Email, Event Name, Position.
- Use test file: Testing/sample_certificates_50.csv.
- Click 'Import CSV' button in Admin Certificates and select the file.
- Select the target Event from the dropdown and submit.

Expected Result:

System parses all rows, validates headers, displays a pre-import preview modal, and issues certificates. Batch record is created with total count, dispatched count, and timestamp.

TC-CSV-02 **Malformed CSV & Duplicate Record Resilience** [ ] PASS [ ] FAIL

Route / Target: `/admin/certificates (Bulk CSV Import Validation)`

Execution Steps:

- Create a test CSV with missing headers, invalid email formats, and duplicate emails for the same event.
- Upload the invalid CSV file.
- Inspect error modal and validation breakdown.

Expected Result:

System rejects invalid rows with row numbers and exact column error reasons. Valid rows are optionally isolated or transaction is rolled back safely without corrupting database.

TC-CSV-03 **One-Click CSV Export of Issued Certificates with Verification URLs** [ ] PASS [ ] FAIL

Route / Target: `/admin/certificates (Export CSV Action)`

Execution Steps:

- In /admin/certificates, apply a filter (e.g., by Event or Delivery Status).
- Click the 'Export CSV' button.
- Open the downloaded CSV file in Excel or Notepad.

Expected Result:

CSV downloads instantly containing columns: Student Name, Email, Event, Position, Certificate ID, Verification URL (qucodex.com/verify-certificate/:id), Delivery Status, Issued Timestamp.

5. Public Certificate Verification Portal & Dynamic Exports

Verifies public accessibility, cryptographic checksum, QR code scanning, and SVG/PDF downloads.

TC-VERIFY-01 **Public Authenticity Verification & Tamper Check** [ ] PASS [ ] FAIL

Route / Target: `http://localhost:5173/verify-certificate/:certificateId`

Execution Steps:

- Open an incognito browser window (logged out).
- Navigate to a valid verification URL: /verify-certificate/<certificateId>.

TC-VERIFY-02 **Live High-Resolution SVG & PDF Download from Verification Page** [ ] PASS [ ] FAIL

Route / Target: `/verify-certificate/:certificateId (Download Actions)`

Execution Steps:

- On the verification page, click 'Download High-Res PDF'.
- Verify PDF generation progress indicator.
- Open downloaded PDF and inspect typography, QR code resolution, and alignment.
- Click 'Download SVG' and verify vector graphic opens in browser.

Expected Result:

PDF and SVG files download cleanly with accurate recipient details, embedded QR code, and official signatures without visual distortion.

6. Brand SEO & Dynamic Meta Title Synchronization

Verifies that all page titles strictly use the official indexed tagline: 'Coding the Future, Today.'

TC-SEO-01 **Root Page & Main Navigation SEO Tagline Verification** [ ] PASS [ ] FAIL

Route / Target: `Frontend Index & Public Pages`

Execution Steps:

- Navigate to http://localhost:5173/ and check browser tab title.
- Inspect HTML source in DevTools (<title>, meta[name='title'], og:title, twitter:title).
- Navigate to /register, /about, /events, /team.
- Check browser tab title on each navigation.

Expected Result:

Titles strictly match: Home -> 'CodeX | Coding the Future, Today.', Register -> 'Join The Club - CodeX | Coding the Future, Today.', About -> 'Coding the Future, Today. - CodeX'. No generic club titles.

TC-SEO-02 **Dynamic Meta Hook (usePageMeta) Verification on Admin & Sub-routes** [ ] PASS [ ] FAIL

Route / Target: `Admin Pages & Dynamic Entity Routes`

Execution Steps:

- Navigate to /admin/login, /admin/dashboard, /admin/certificates.
- Inspect the browser tab title as routes change without page reload.
- Open /verify-certificate/:id and wait for certificate data to resolve.

Expected Result:

usePageMeta hook dynamically synchronizes document.title and meta tags to '<Page> - CodeX | Coding the Future, Today.'. Certificate verification dynamically displays recipient name and position.

7. Database Performance & Safe Index Management

Verifies compound indexes in Mongoose models and cleanup of redundant migration scripts.

TC-DB-01 **Mongoose Schema Compound Indexes & Query Execution Speed** [ ] PASS [ ] FAIL

Route / Target: `Backend MongoDB Certificate & CertificateBatch Models`

Execution Steps:

- In MongoDB Compass or mongosh shell, connect to the CodeX database.
- Run db.certificates.getIndexes() and db.certificatebatches.getIndexes().
- Verify compound indexes exist for: { event: 1, email: 1 } and { certificateId: 1 }.
- Confirm redundant script Backend/src/utils/optimizeDbIndexes.js has been safely eliminated.

Expected Result:

Queries on certificateId and event+email execute with ColScan: false (using B-tree index scan), ensuring sub-50ms lookup times under production loads.

8. Test Execution Summary & Quality Sign-Off Matrix

Official sign-off sheet for QA Engineer, Tech Lead, and Release Manager prior to production merge.

TEST SUITE SUMMARY & STATUS TRACKER			
TS-01: Admin Authentication & Timing Attack Defense	TC-AUTH-01 to TC-AUTH-04	4 Cases	**[ ] PASS [ ] FAIL**
TS-02: Rate Limiting & HTTP 429 Cooldown Telemetry	TC-RATE-01 to TC-RATE-03	3 Cases	**[ ] PASS [ ] FAIL**
TS-03: Certificate Rejection Deletion & Edit Sync	TC-CERT-01 to TC-CERT-03	3 Cases	**[ ] PASS [ ] FAIL**
TS-04: Bulk CSV Import, Validation & Export Flow	TC-CSV-01 to TC-CSV-03	3 Cases	**[ ] PASS [ ] FAIL**
TS-05: Public Certificate Verification Portal & Downloads	TC-VERIFY-01 to TC-VERIFY-02	2 Cases	**[ ] PASS [ ] FAIL**
TS-06: Brand SEO & Tagline Metadata Synchronization	TC-SEO-01 to TC-SEO-02	2 Cases	**[ ] PASS [ ] FAIL**
TS-07: Database Schema Indexes & Safe Script Cleanup	TC-DB-01	1 Case	**[ ] PASS [ ] FAIL**

DEFECT REPORTING & SEVERITY TRIAGE GUIDELINES

- Blocker (P0): Authentication failure, bypass of MFA lockout, or data corruption in certificates/DB.
- Critical (P1): Rate limiter fails to trigger 429, CSV upload crashes server, or certificate fails to delete on rejection.
- Major (P2): Cooldown banner timer mismatch, CSV export missing columns, or PDF download visual misalignment.
- Minor (P3): Title tag typo, minor padding/margin inconsistency, or non-blocking console warnings.

Report all defects on GitHub Issues under branch 'feat/super-admin-and-production-certificates' with reproduction steps.

OFFICIAL STAKEHOLDER SIGN-OFF MATRIX				
Lead QA Test Engineer:	_____	**Date:**	____ / ____ / 2026	**Verdict: [ ] APPROVED [ ] REJECTED**
Technical Lead / Super Admin:	_____	**Date:**	____ / ____ / 2026	**Verdict: [ ] APPROVED [ ] REJECTED**
Release Management:	_____	**Date:**	____ / ____ / 2026	**Verdict: [ ] MERGE READY**